Bruker microCT

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MTS2 user guide

V2.1 20/10/2015

Index

1. Item list

Below an image of the items shipped with the Material Testing Stage version 2 are shown.

1. Material testing stage with dust protection cap for transportation

2. Box (left) containing

    a. Large tensile sample holder
    b. Large top and bottom clamp with tool
    c. Base ring

3. Holder closing ring

4. Box (right) containing

    a. 3 compression platforms
    b. 1 spacer for the compression holder
    c. compression mode holder
    d. platform contra nut
    e. large compression mode holder

5. Box (top left) containing

    a. top part of tension mode holder
    b. bottom part of tension mode holder
    c. spacer for high resolution compression holder
    d. high resolution compression holder

6. CD with control software

7. Emergency release key

8. Hexagonal screwdriver for tension sample mounting


5
a
b
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7
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2
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e

2. Software installation

The supplied CD contains the control program of the material testing stage entitled “MTS2.exe” and settings file named “MTS2.ini”. Copy both files into the directory with the other Bruker microCT software.

If you don’t know which directory this is, you can find it by right clicking on the screen shortcut of the Bruker microCT control software and select “Open File Location”.

In order to have a quick start of the stage control software it is advised to create a shortcut on the desktop by right clicking the file and selecting “send to desktop”.

Software updates are posted on the website at

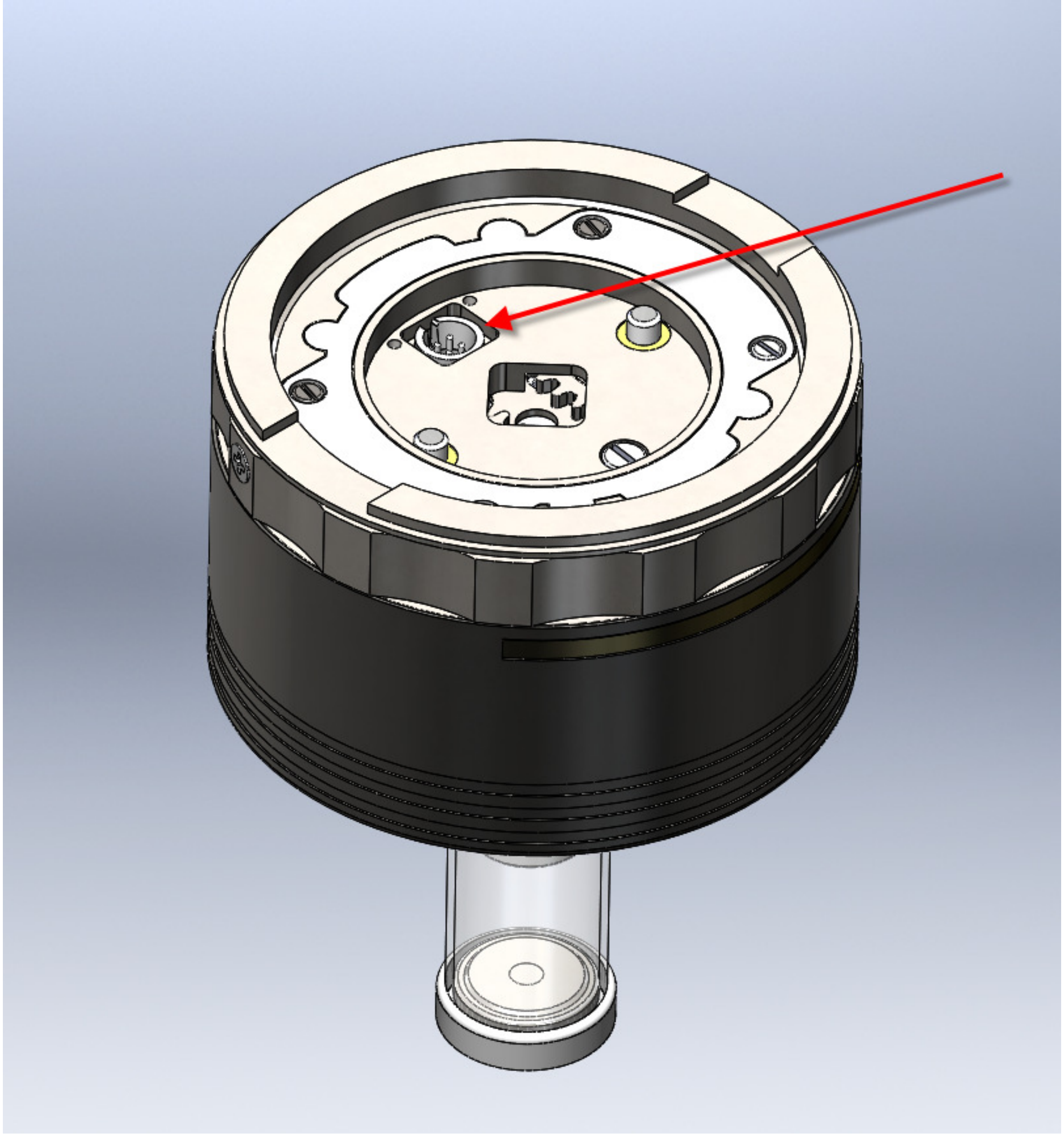
http://www.skyscan.be/products/downloads.htm

3. Mounting the stage in the scanner

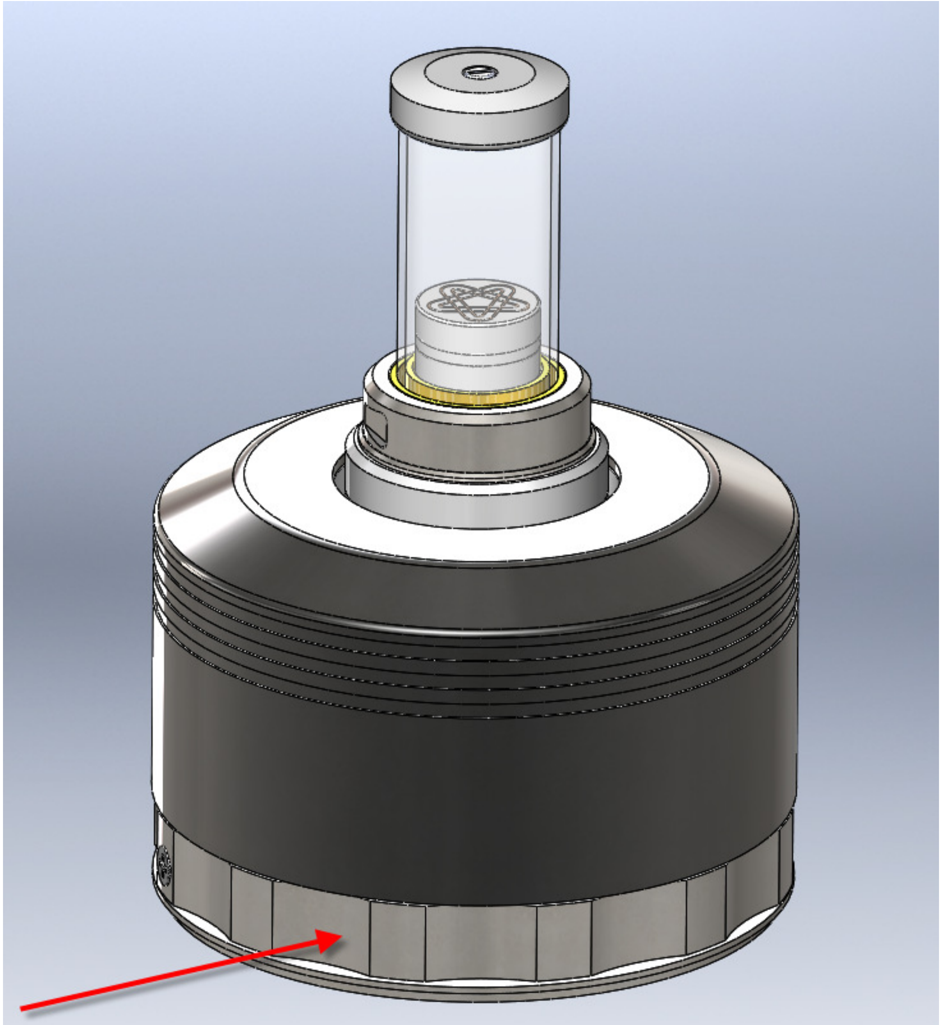
To install the stage in the microCT scanner, start the scanner and open the specimen chamber using the scanner’s control software. Unscrew and remove the standard brass cylindrical holder on the top plate of the scanner’s specimen stage.

Align the four pin connector (see image below) with the matching connector of the scanner. Gently lower the material testing stage until the connector snaps in place.

! When adjusting the magnification and vertical positioning using the scanner stage control be mindful not to come into contact with the plate on the source side. The maximum magnification using the different testing tubes is dependent on the scanner version. The Visual Camera can be used as an additional safety guide. It is advisable to leave the MTS control software running to restrict the vertical travel limit of the scanner stage.



Secure the MTS to the stage by rotating the attachment wheel (see image below) clockwise until finger-tight.



4. Compression testing

Depending on the width of the sample select one of the 3 platforms provided with the system, screw the platform in the spacer and choose the correct height. Place it in the central top opening of the MTS with the open part down. Place the sample on top of the spacer. Put the compression mode tube (image below, bottom right item) over the sample; align the three brass feet of the tube with the matching gaps on the MTS. Use the holder closing ring (image below, top right item) to attach the tube to the MTS; rotate clockwise until finger-tight.

! Warning: Improper mounting can lead to damage to the MTS stage.

The system is designed in such a manner that the central zone of the sample will stay static during compression. Although the test can be started at any position of the stage, it is advisable to lower the specimen holder before compressing. This can be done by switching to tensile mode and running the motor until it reaches the bottom position.

It is advised to put the sample in the center of the disc, if possible. In order to have the best performance, the load should be applied parallel to the main axis of the load cell. This ensures no cosine error is induced. It is also advisable to use samples with flat top and bottom surfaces.

Although the mechanical construction aims to minimize any inclination of the sample holder, centering the sample will improve the accuracy of the measurement.

Ideally there should be a slight gap between the top of the compression tube and the specimen so the load cell does not register any load at the beginning of the test.

For demounting the stage the inverse of the described procedure is advised. It can occur that the closing ring has tightened beyond finger-tight during testing. In this event the emergency release key (see image below) can be used to unscrew it. Note that this key is not intended for sample mounting.

5. High resolution compression testing

The procedure for mounting the high resolution compression testing stage for small samples is similar to that for normal compression testing (see section above). The tube and spacer are shown below, indicated by the red oval..

6. Tensile testing

To use the MTS in tensile mode we will use the tube and base which can be seen on the left side of the picture above. First insert the bottom part of the stage in to the top of the MTS. Twist the stage in either direction by 60° to secure it, a click can be felt when doing so. When done correctly it will no longer be possible to lift the component out.

Slide the top part of the tensile holder, i.e. the tube, over the bottom. One can get access to the interior of the tube by removing the top cap and front plexi window of the tube. It can be of interest to secure the sample to the top part of the stage before sliding it over the bottom. Secure the sample tightly to the top and bottom using the provided hexagonal screwdriver (see image below).

One can preposition the displacement to be suitable for the length of the sample by using the control software as described in section *7.i*. By turning the metal ring (indicated by the red arrow in the image above), the displacement can still be adjusted further slightly.

Put the plexi front of the tube back in place as well as the top cap (see image below).

Use the holder closing ring to attach the tube to the MTS; rotate clockwise until finger-tight.

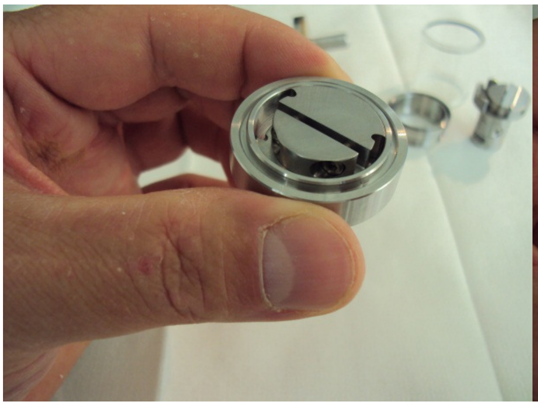
! Warning: Improper mounting can lead to damage of the MTS stage.

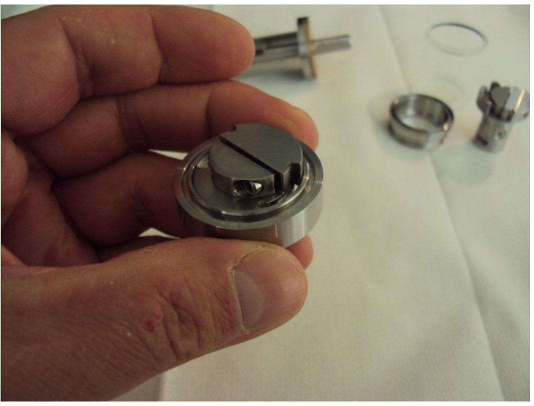
As discussed in section 4, if the closing ring has tightened beyond finger-tight during testing, the emergency release key can be used to unscrew it.

7. Large Tensile testing

The sample holder consists out of 4 key pieces, from left to right: top clamp, sample tube, bottom clamp, and base.

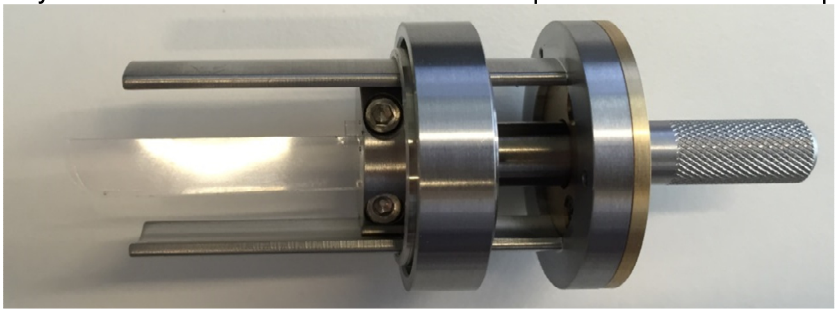
The top clamp can be assembled in two different positions.
You can choose the most convenient for you. This depends on the sample length and actual height position from the MTS.





Position 1 Position 2

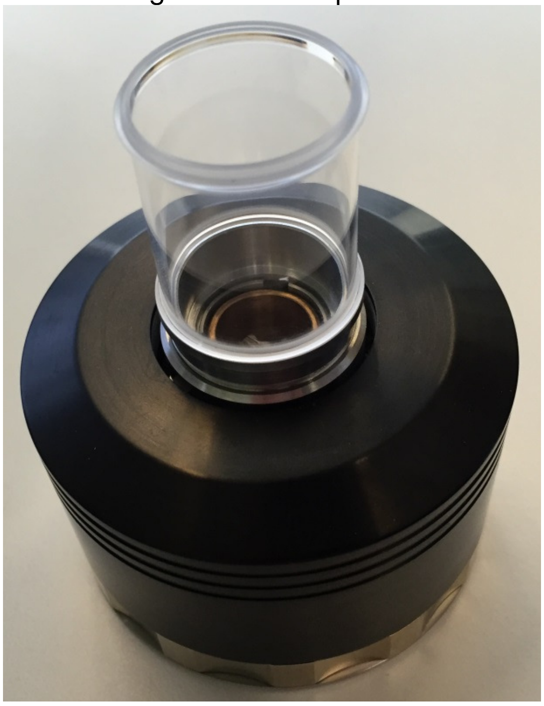
You can put the sample in the top clamp and secure the sample in the top clamp by tighten the three screws Shove the clamp over the tool. Make sure the holes match so you can insert the tool to hold the sample like showed in the picture



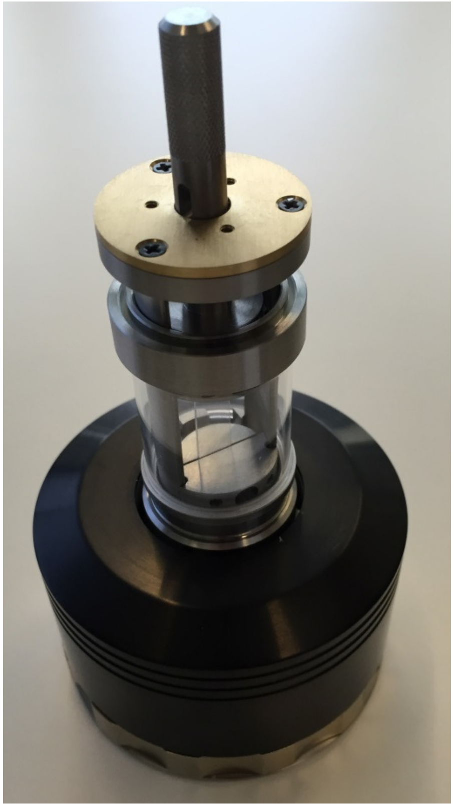
below.

At this point you can add the bottom clamp. Shove the sample to the top and add the bottom clamp to the tool. Now slide the sample between the clamps of the bottom part and secure the sample

Before mounting the assembly on the MTS stage we need to prepare the MTS stage by installing the lock ring and the sample tube holder



When the MTS stage is prepared you can mount the whole sample assembly on the MTS. Make sure the assembly is installed well.
Now turn the handle of the tool to lock it to the MTS and retract the tool



For more large sample mounting please watch the video’s on the delivered disk in the Tensile directory

8. Control software

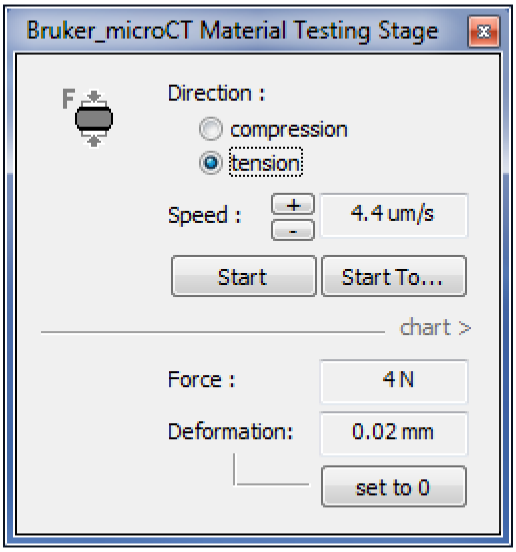
The control software is used to manipulate the stage and to display its sensors' output in real time. This section describes all settings, options and available user modes.

The goal is to register both the load applied to the object and the resulting displacement and to show this on screen in a graph. This allows the user to make microCT scans of the sample at a predefined load or displacement and to investigate the results.

The software will only start when the testing stage has been put in the scanner and the connection to the computer has been made.

i. Main functions

The user first has to select the direction in which the stage will move, i.e. compression or tension. The speed of movement can be adjusted, the maximum limit being 4.0um/s.



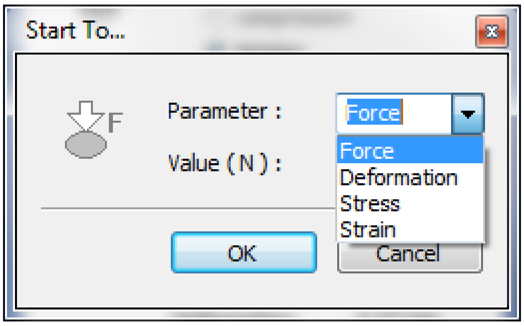
There are 2 user modes indicated by "Start" and "Start To…":

- **Start**

In this mode the stage will move in – compress – or out – tension – until the maximum allowable load is registered or until the maximum displacement is reached.

The user can start or stop the motion at any time using the [Start/Stop] button.

- **Start To…**



In this mode the stage will move to a predefined load, deformation, stress or strain.

The motion will be stopped any time the maximum allowable load or the maximum displacement has been reached.

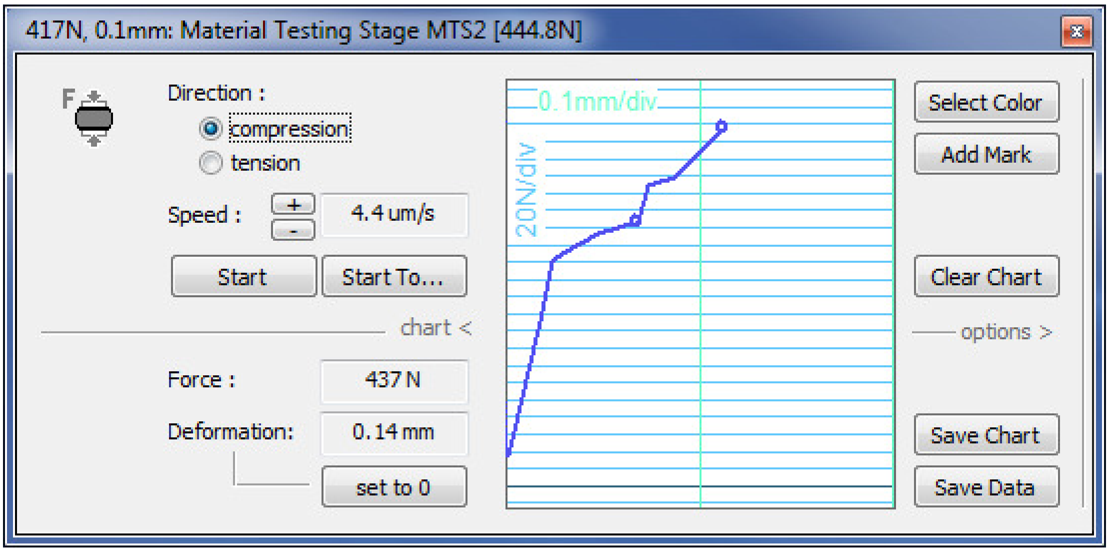
The user can start or stop the motor at any time using the start/stop button.

To get the maximal stroke length for compression, the stage should be moved to maximum extension first, for tensile testing it is the inverse. To do so either the "Start" or "Start To..." options can be used. The test can be started at any position if the maximum travel is not needed.

ii. Data visualization and saving

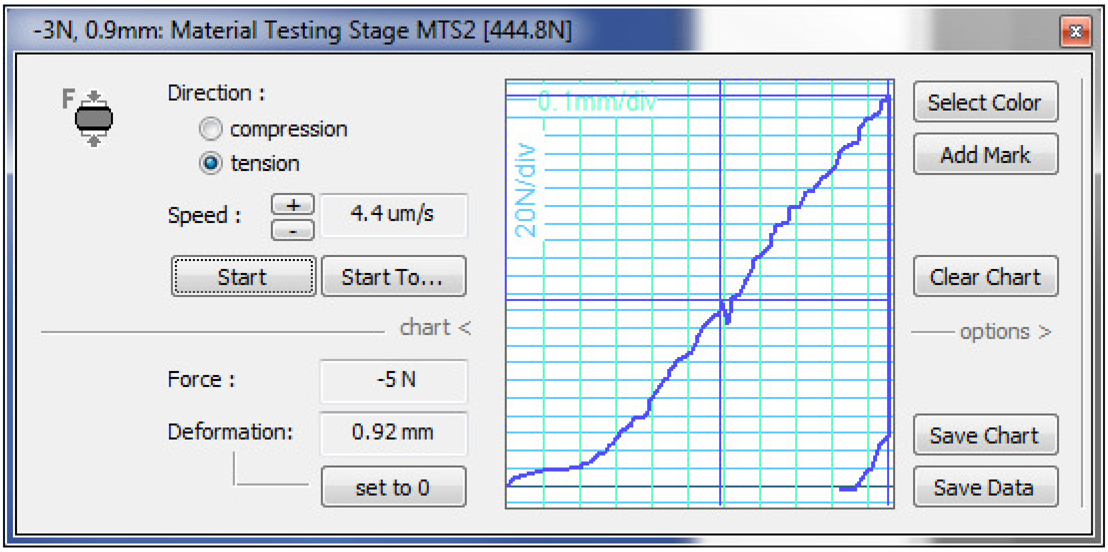
While running, the software will register either the stress and strain or the force and deformation relative to the 0 position. The default 0 position is the initial starting position, which can be adjusted by clicking the "set to 0" button.

Clicking the **"chart >"** expands the pane with the graph of the results. The range of both X- and Y axis will be resized automatically depending on the output of the sensors. While the load is below the sensitivity of the load cell, the graph will not be plotted; this situation is common at the start of a new test cycle.

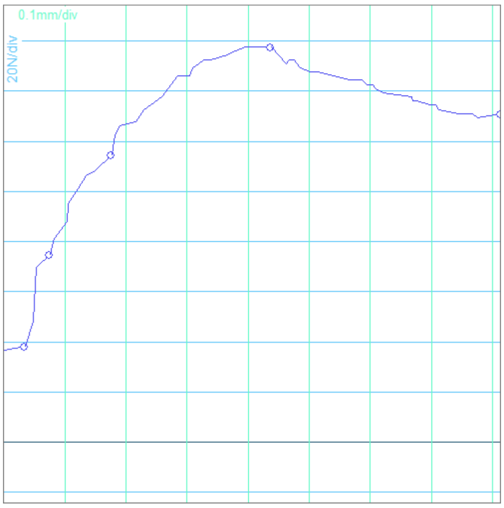


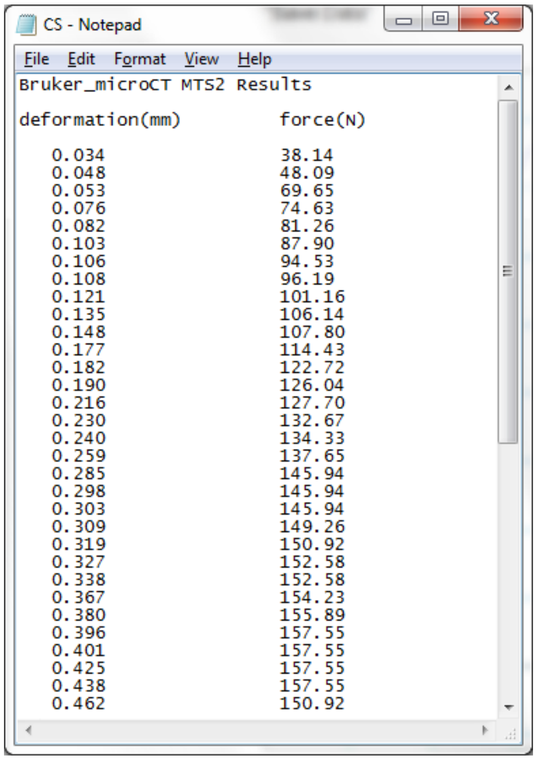
The graph can be changed in color, marks can be added or the chart can be cleared by clicking the respective buttons on the chart pane. For the marks there is the choice between crossed lines or rounds, for information on how to change between these we refer to section *7.iv*.

Inversing the testing direction allows the elastic properties of the material to be evaluated.



The “**Save Chart**” button allows the chart to be saved in bitmap format. “**Save Data**” allows for a txt-file to be generated of said information.

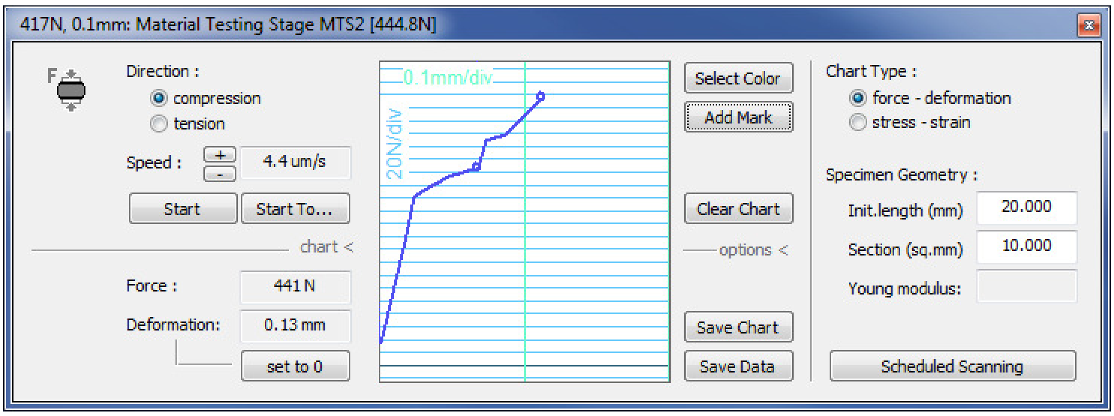




Bruker_microCT MTS2 Results

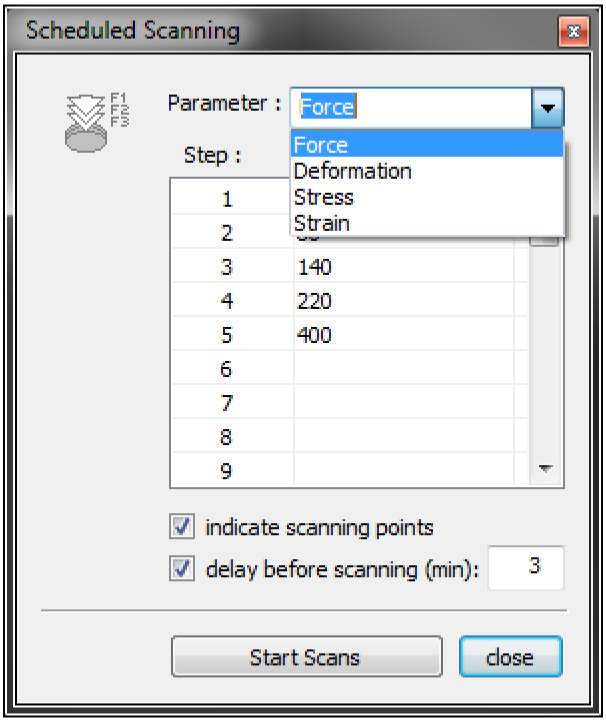
deformation(mm)	force(N)
0.034	38.14
0.048	48.09
0.053	69.65
0.076	74.63
0.082	81.26
0.103	87.90
0.106	94.53
0.108	96.19
0.121	101.16
0.135	106.14
0.148	107.80
0.177	114.43
0.182	122.72
0.190	126.04
0.216	127.70
0.230	132.67
0.240	134.33
0.259	137.65
0.285	145.94
0.298	145.94
0.303	145.94
0.309	149.26
0.319	150.92
0.327	152.58
0.338	152.58
0.367	154.23
0.380	155.89
0.396	157.55
0.401	157.55
0.425	157.55
0.438	157.55
0.462	150.92

By clicking “**options >**” the pane is expanded allowing the chart type to be changed to the stress-strain type. In order for this data to be accurate the specimen geometry – initial length and cross sectional area - needs to be entered in the respective boxes. Additionally an indication of the Young modulus is displayed here.



iii. Scheduled Scanning

Scheduled Scanning is used for automatic multiple scanning at different load or displacement settings. One can access the menu by clicking the "Scheduled Scanning" button found in the options pane.

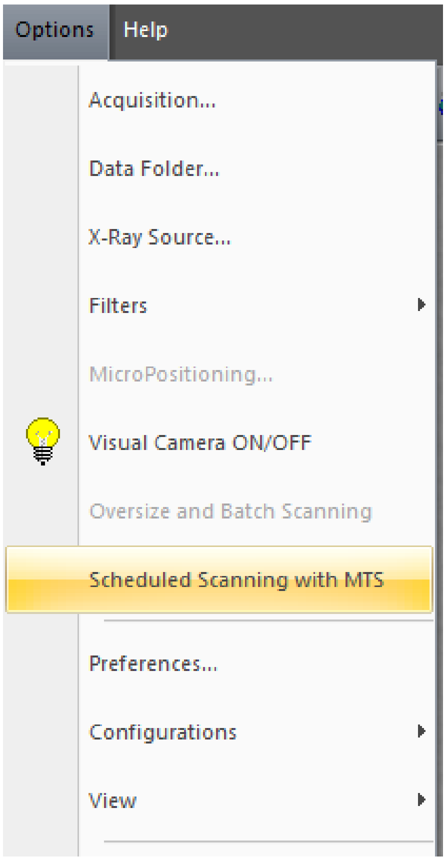


First make selection of the parameter, i.e. either force, deformation, stress or strain.

Values for the different steps can be entered in the relevant table.

By checking the **"indicate scanning points"** option marks will be added to the graph automatically at the respective points.

A relaxation time can be set by checking the **"delay before scanning (min):"** box and specifying the holding time in minutes. Note that during operation the sample may heat up and that this can have an influence on the mechanical properties of the sample.

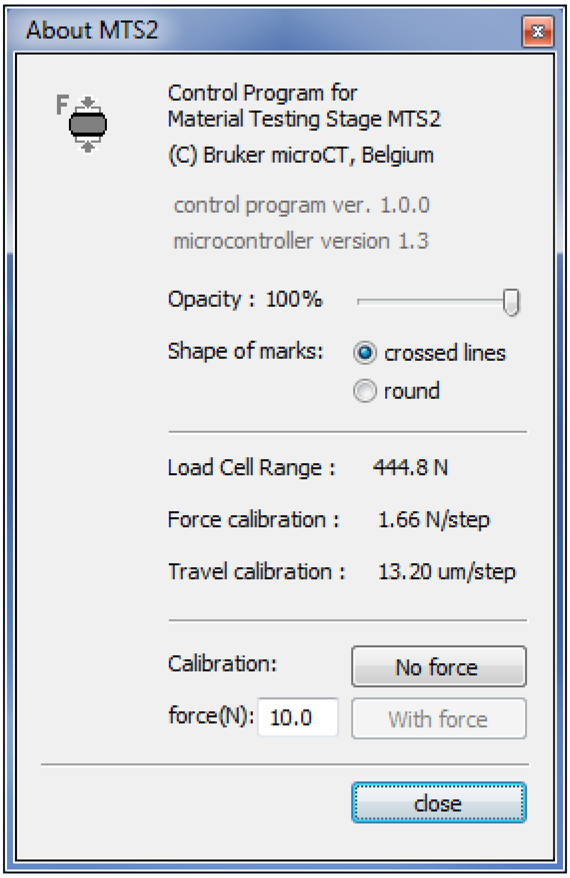


The batch can be launched from the scanner control software by selecting "Scheduled Scanning with MTS" from the options menu with the following selection of "Start Scans in the MTS program"

Scan setup is similar to usual practice; it is advisable to center the FOV on the middle of the test sample as this will remain static during testing. The different scans will have a suffix '…~#NNMM~LLLLLLLL' where NN denotes the step number indexed from 00, MM indicates the total number of steps and the final eight L's reflect the indexing of angular positions during scanning (as for the ordinary scan)The reconstruction software NRecon and visualization software DataViewer recognize this format automatically and will be able to reconstruct and display them with the possibility to scroll over the selected set of forces or displacements.

iv. Settings and calibration

The "About MTS2" menu can be opened by right-clicking on the top bar of the MTS control software and selecting the relevant button. The versions of control program and microcontroller can be verified here.



The opacity of the control software pane can be changed using the slider. The shape of the marks on the graph can be changed from crossed lines to round here as well.

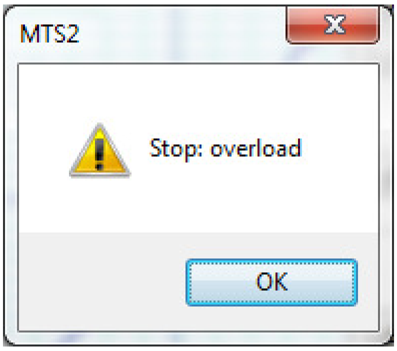
The load cell range, force and travel calibration can be found here. The calibration factors are determined by the manufacturer; these settings are fixed and cannot be modified by the user.

Operating environment and transducer history can change the calibration balance of the MTS. These factors include temperature, transducer age and transducer overload. The zero offset can be reset by selecting the **"No force"** button when no load is applied. Recalibration of the load cell can be done in two steps. Firstly click the "**No force**" button when no force is applied on the bottom of the stage. Then apply a well-known force by putting a measured weight on the central rod of the MTS, enter the applied force into the editable field after "force (N)" and press the "**With force**" button

! Warning: Improper calibration will lead to biased results.

9. Integrated safety precautions

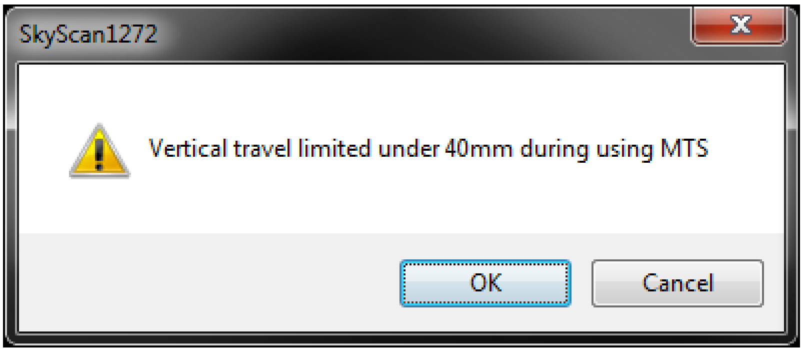
To protect the drivetrain and the load cell of the testing stage some limitations have been built in. Due to these limitations the stage may come to a stop not expected by the user. The motor will stop once maximum displacement has been reached.



The motor will stop once the maximum load has been reached and give the message **"Stop: overload"**. This is the overload protection of the load cell. When a load cell's maximum designed input limit is exceeded it might be damaged. The value is dependent on the type of load cell installed and cannot be modified.

! Notice that the overload protection only works during normal operation. Be careful not to drop any object on the stage when the tube has been removed. Exposing the internal load cell to fast acting loads may result in damaging the sensor.

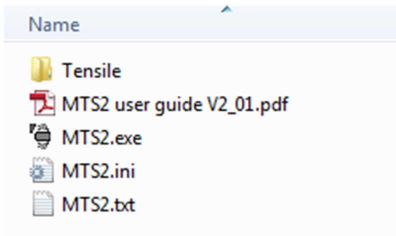
When the MTS control software is running the vertical travel limit of the microCT stage will be redistricted to protect the MTS and the scanner from damage.



! When positioning the MTS and the sample for scanning it is advised to leave the MTS control software running to restrict the vertical travel limit.

10. Two MTS stages on same scanner

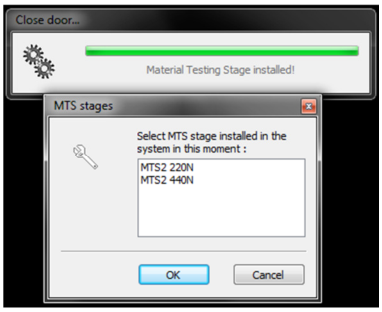
When using 2 different MTS stages on the same scanner there are some things that need to be considered. In case the MTS software is not preinstalled on the control PC



By default the stages are delivered with a small DVD where you find the “MTS2..exe” and “MTS2.ini”. The .ini file contains the calibration values for the MTS and is unique for each MTS.
Change the names of the MTS .exe and .ini file . The .exe and corresponding .ini file must have the same name.
For example: MTS2_440N.exe and MTS2_440N.ini

Then you need to copy these files to the Skyscan directory where the Skyscan control software is located.
The MTS software needs to be in the same folder than the Skyscan control software.
Otherwise you will get a time out error when doing scheduled scanning.

When using 2 MTS stages there is a small difference when a MTS is installed, the control software will ask which MTS is installed after you installed or switched the MTS stage or started the scanner with an MTS inside.



When above window pops up, choose the right MTS stage that is installed. The stage will use the calibration value of the one you selected. In case you selected the wrong stage, the stage will use wrong calibration values and will not display correct values.

11. General guidelines

- The stage should be stored at room temperature in a dry environment. The dust protective cap does not need to be placed back for short term storage; it is primarily intended for transportation protection.
- Release the load prior to storing, it is advised not to keep any load on the load cell before demounting.
- Do not attempt to operate the stage outside the Bruker micro CT scanner.

Our customer's opinion is appreciated. Suggestions or comments to improve the device or control software are welcome at applications@bruker-microCT.com.